

Deciphering Archaeological Spatial Structure on Domestic Sites

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Archaeological Spatial Structure

Two components

- “*Site Structure*”: the spatial distribution of subsurface features across sites (*e.g.* remains of structures, fences, pits, etc).
- *Artifact Distributions*: the spatial distribution of artifacts and their traces across sites.

History

- Serious interest in spatial structure is relatively recent.
- *Culture History*. Primary focus: The distribution of “cultures” in time and space. Sites are "homogeneous artifact mines"
- *New or Processual Archaeologists (1960's)*. How cultures function is important too, *e.g.* social and social organization. Intra-site spatial structure is important!

Archaeological Spatial Structure

Deciphering archaeological spatial structure: making inferences about the causal dynamics responsible for the spatial patterns .

We need to think about two sets of processes.

Systemic context processes

- Site structure: what factors affect the spatial organization of activities and the facilities (whose archaeological traces are “features”) with which activities are associated?
- Artifact Distributions: what factors affect where artifacts are discarded?

Archaeological context processes

- Site Structure: What factors affect the ways in which facilities are represented as the features we see in the record?
- Artifact Distributions: What factors affect spatial patterns after artifact deposition and site abandonment?

Systemic context processes

Spatial organization of activities and the facilities with which they are associated: a simple model.

The extent to which one activity will be spatially segregated from others is a function of

- Benefits of moving
 - Scale inversely with interference potential:
 - activities take up lots of space
 - activities are dangerous
 - activities require uninterrupted time
 - *Higher interference potential means activities are more likely to be moved farther away.*
- Costs of moving the interfering activity
 - Scale with distance moved, access frequency.

General vs. Special Activity Areas

General Activity Areas

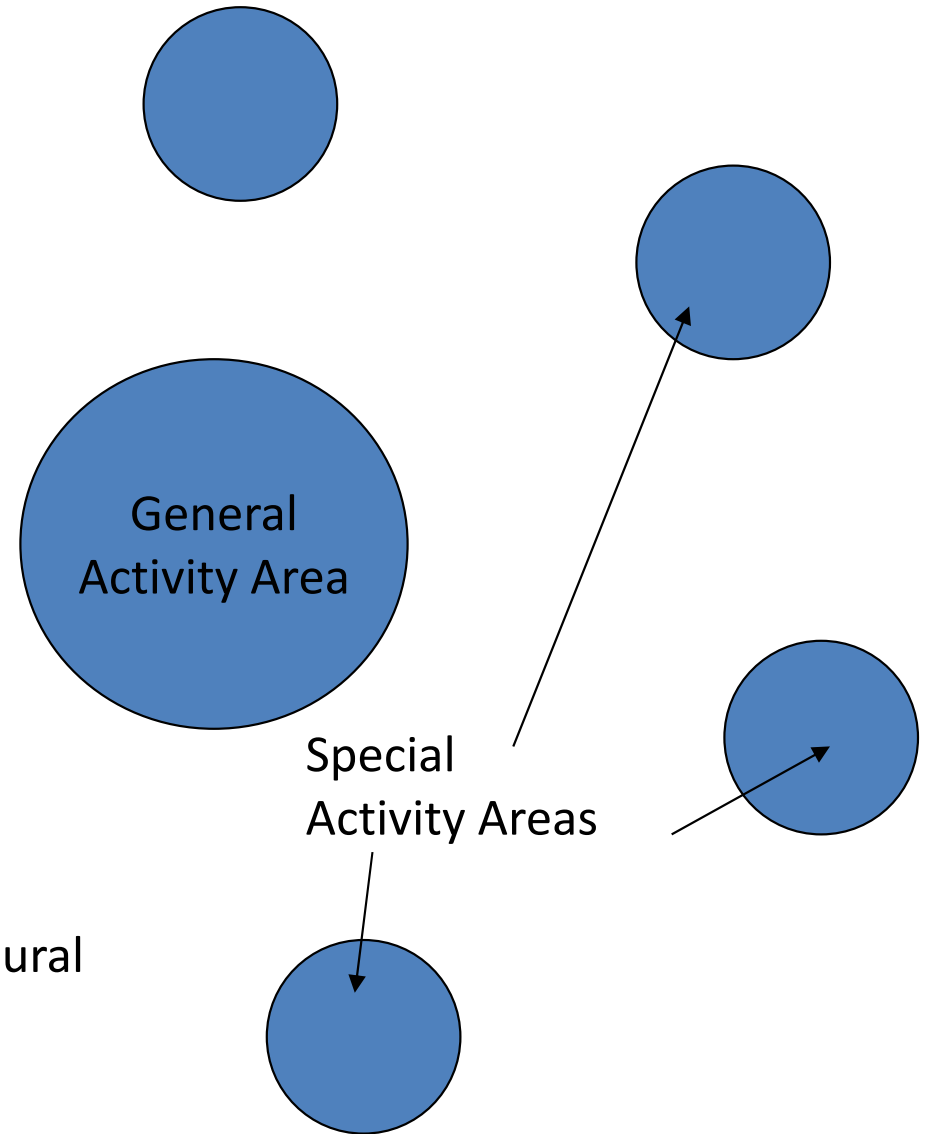
- sleeping, eating, socializing, cooking
- sleeping, cooking often separate within them

- Family (kin based)
- Communal (cross cut kin lines e.g. gender based)

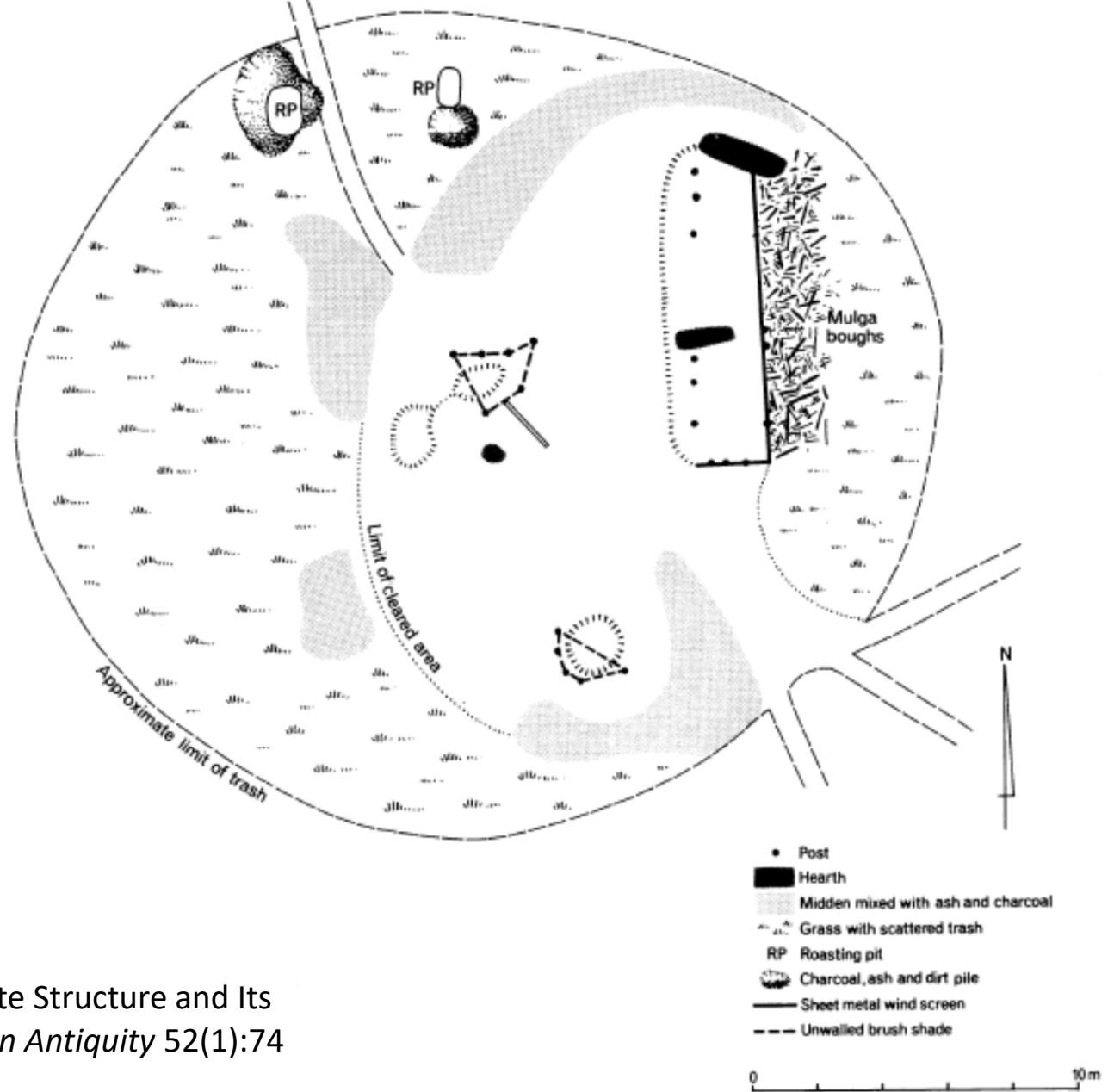
a.k.a, “nuclear” (John Yellen)
“intensive” (Lewis Binford)
“household” (O’Connell)

Special Activity Areas

- ecologically variable
- e.g. kangaroo roasting, car repair
- cooking for large numbers (signals)
- bulk processing and storage of agricultural produce
- lots of refuse, space, time



Implications for Site Structure



Source:

O'Connell, James, 1987, "Alyawara Site Structure and Its Archaeological Implications. *American Antiquity* 52(1):74

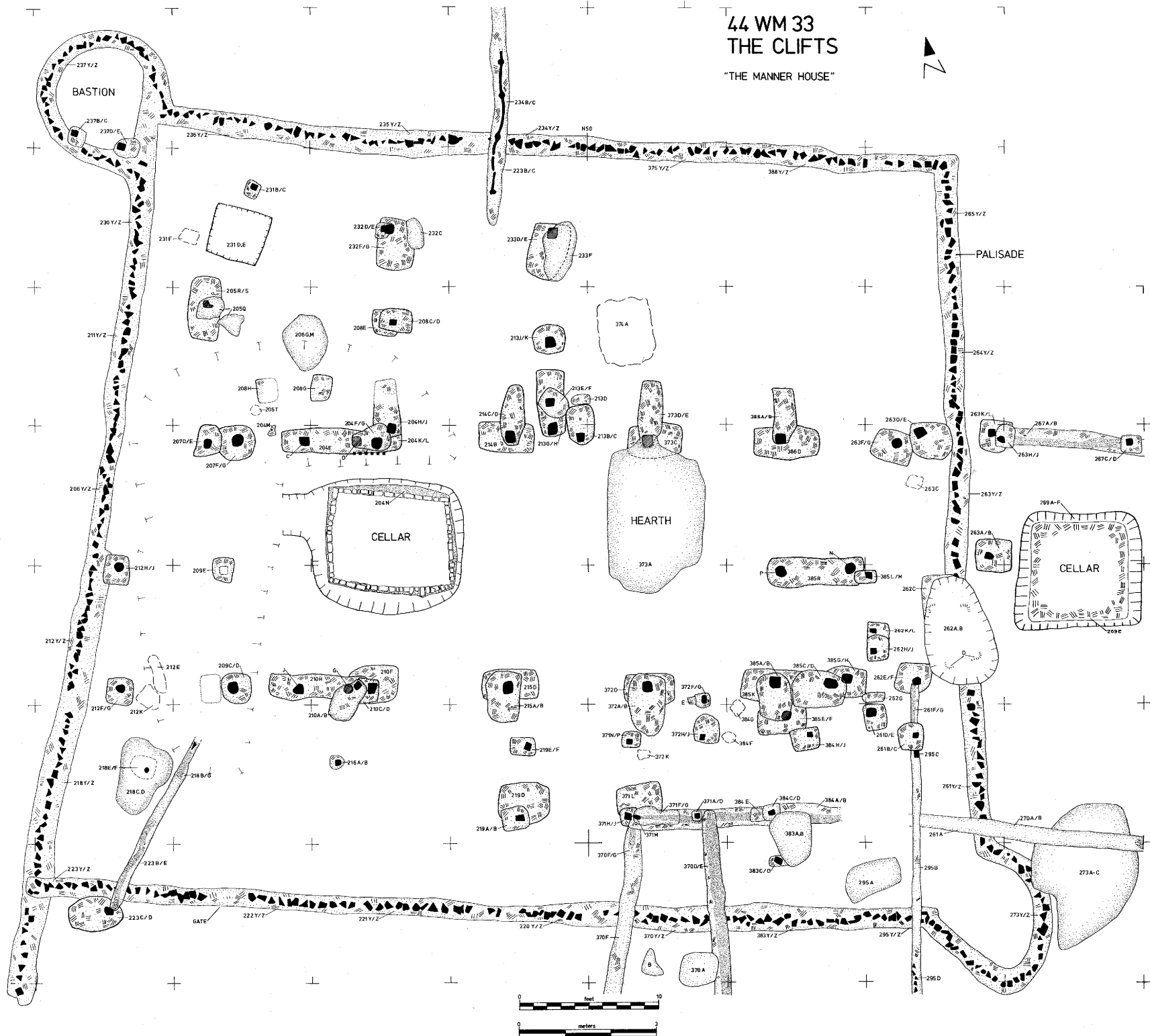
Figure 1. Plan of Alyawara household activity area, adjacent roasting pits, and refuse disposal zone.

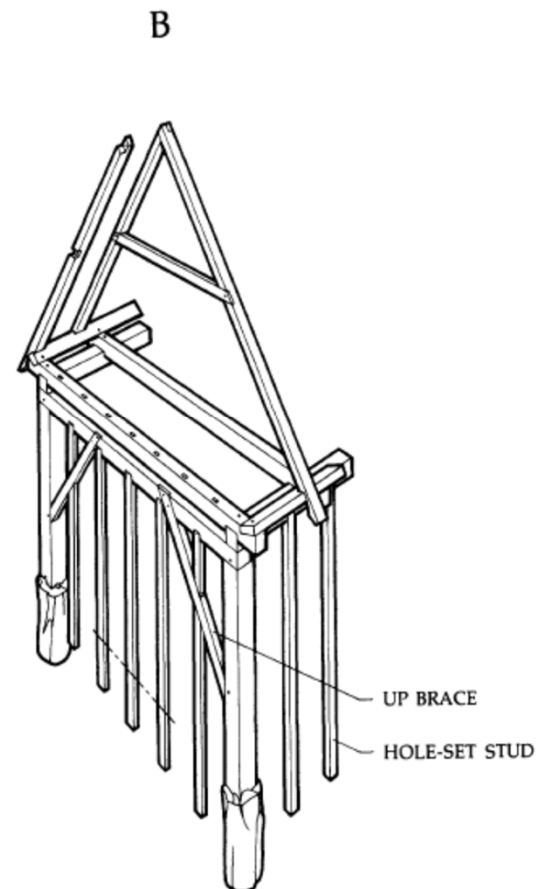
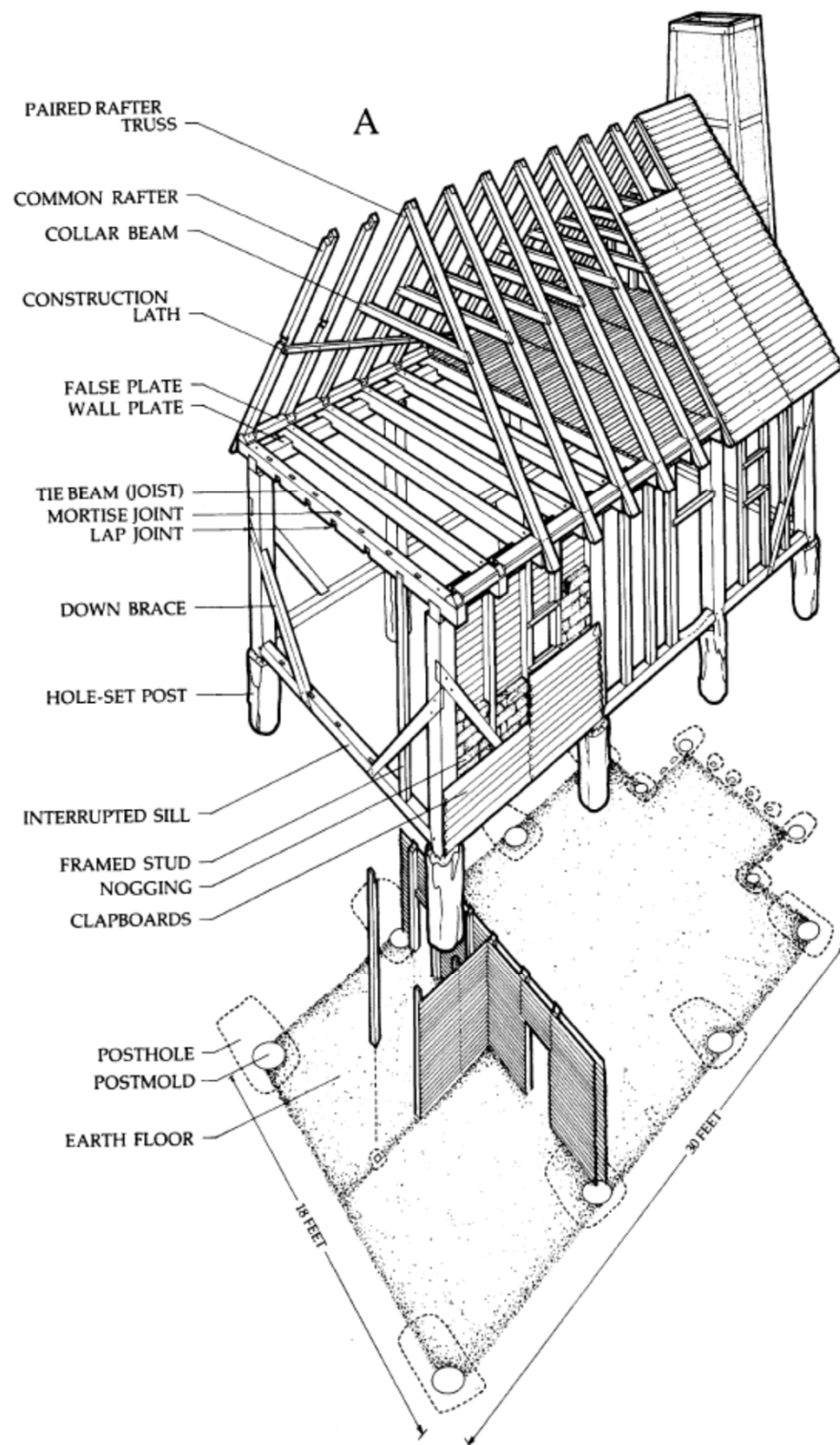






"THE MANNER HOUSE"









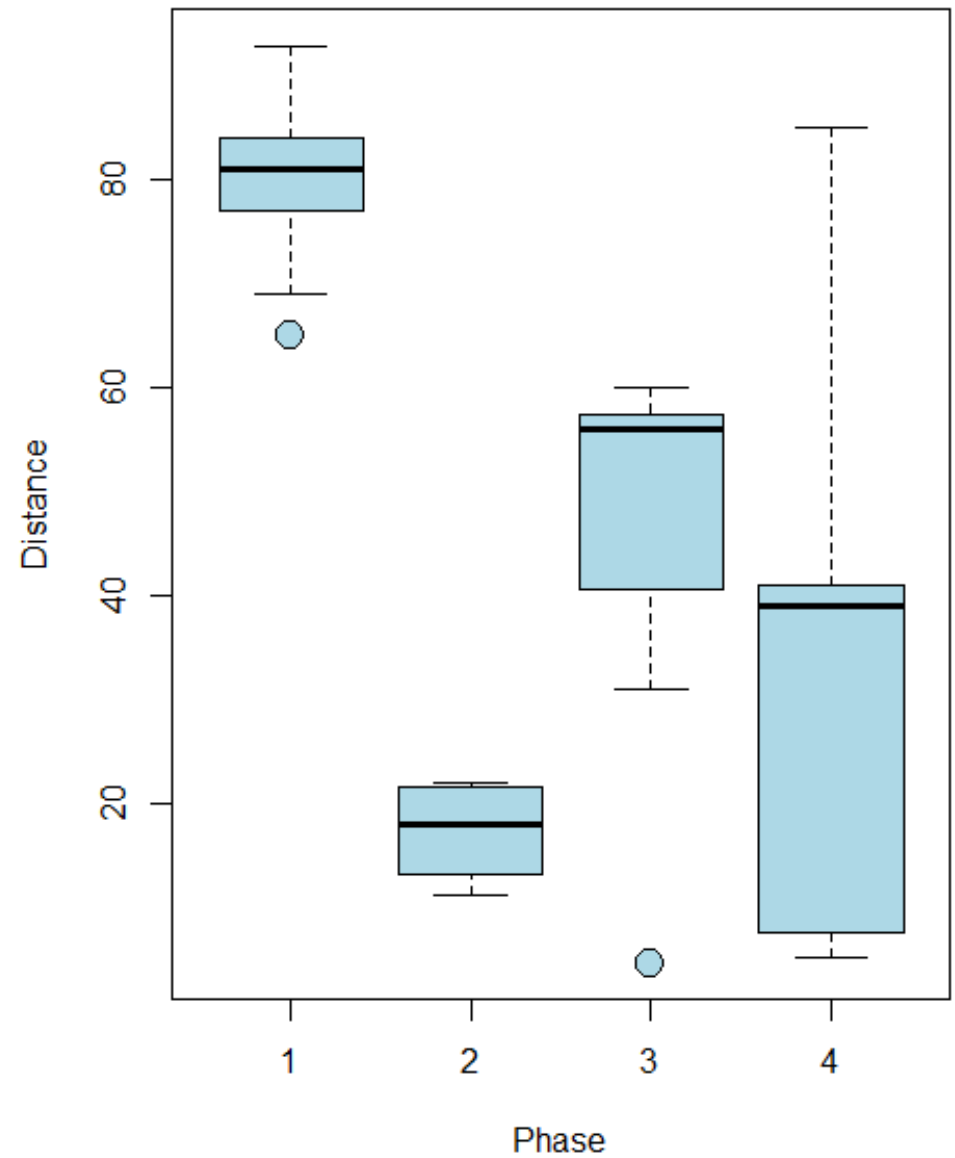
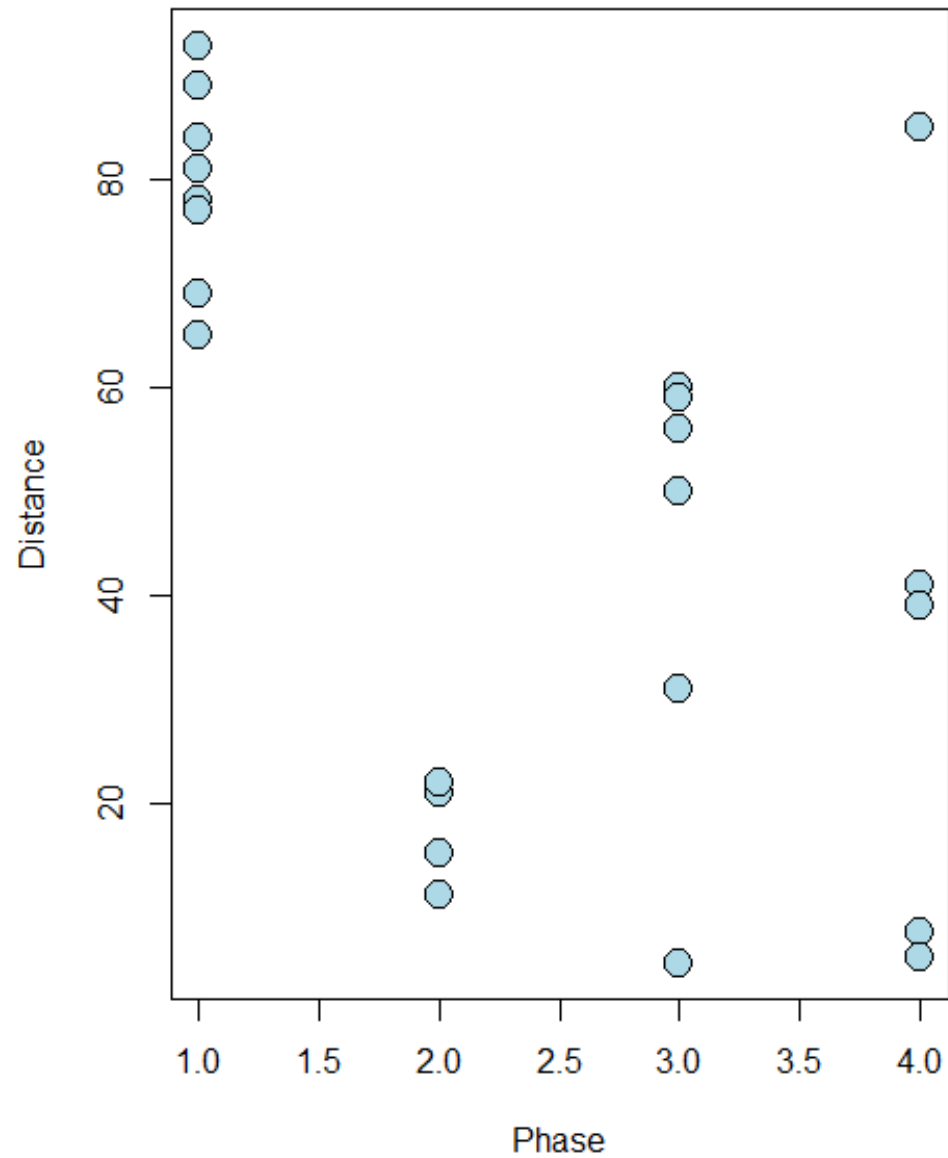




Context	Distance	Length	Width	Depth	Phase	
305A_G	78		10.7	9.7	1.8	1
304A_C	93		7.3	3.5	1.1	1
346A_D_362A_C	84		8.2	5	0.9	1
351B_H	69		10.2	3	1.3	1
356A_C	77		3.7	1.5	0.3	1
390M_N	65		6.2	4.2	0.8	1
360A	81		2	2	0.3	1
290A_D	89		3.5	3	0.6	1
289A_H	84		10.5	5.5	1.7	1
250D_E	21		7.5	6.4	0.7	2
273A_C	15		10	8	1.2	2
274A_B_E_G	22		11	7	1.2	2
295A	11		4.1	2.3	1	2
205G_M	4.5		4	3.1	0.6	3
240F_G	31		4.5	3.1	0.6	3
259A_D	56		5.5	4.5	0.7	3
288C_R	60		12	8.5	1.3	3
288S_AD	59		13.5	4.4	1.2	3
255A_E_Z_AH	50		13.5	10.5	1.1	3
255F_Y	56		5.6	4.5	2	3
309A	56		5.5	5	0.6	3
226B	7.5		4	3	0.3	4
277A_C	41		11.5	4.7	0.5	4
280A_H	39		10	7.4	1.1	4
383A_B	5		3.9	3	0.5	4
345A_D	85		5.4	4.4	1.1	4

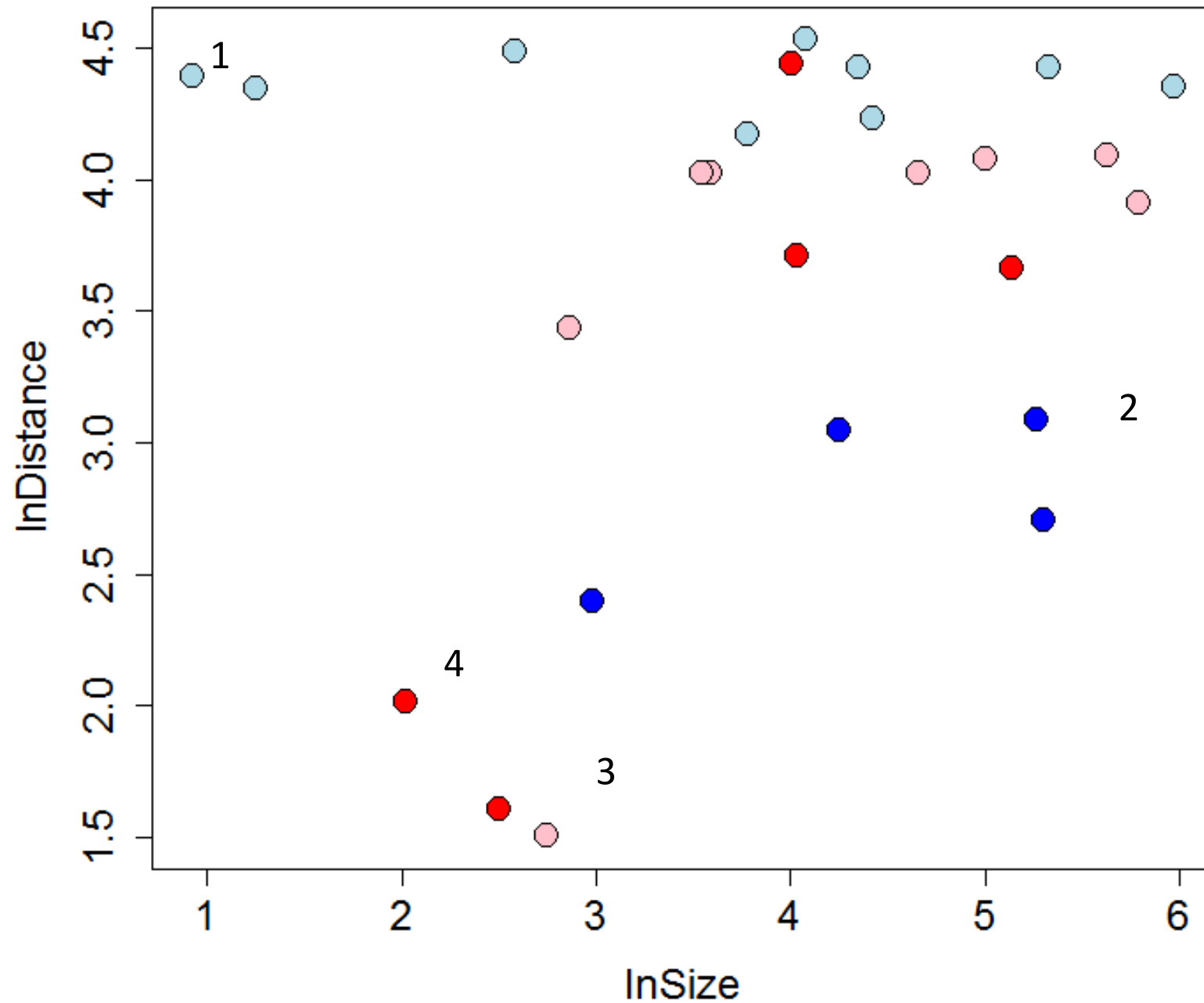
Clifts Plantation

Distances of pits from the main house



Clifts Plantation

Pit Distances vs. Pit Size (log scale)

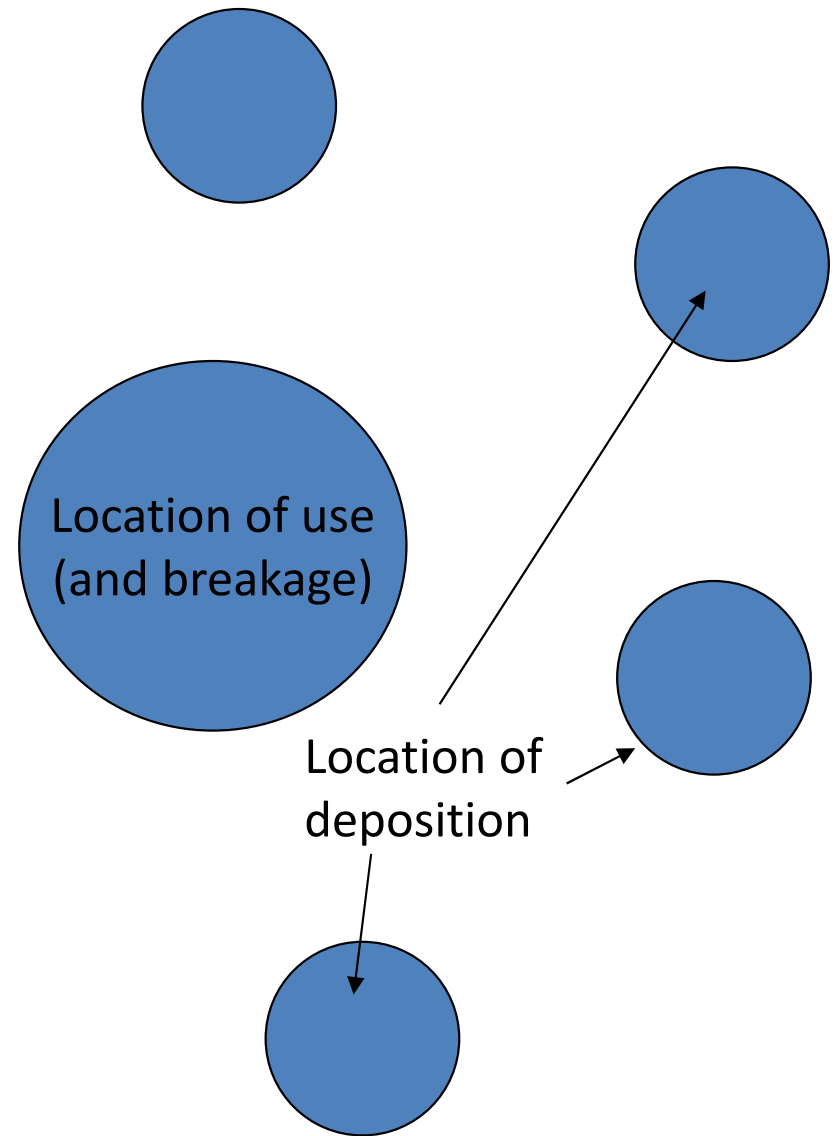




Systemic context processes

Spatial organization discard

- ***The Pompeii Premise***: the *false* idea that the location of artifact discard is the location of use and breakage.
- ***Primary refuse*** vs. ***secondary refuse*** (Mike Schiffer)
- *Cleaning up* creates some patterns and destroys others
- How far broken/failed artifacts are transported effects spatial scale at which spatial patterns can be interpreted.
- ***Secondary refuse nearly always has useful spatial pattern at some informative scale!!***



Systemic context processes

Spatial Organization of Discard.

The distance that artifacts will be transported from the location of use and breakage is a function of...

- Benefits of cleaning up
 - Scale inversely with interference potential with other activities
 - Amount of artifacts $\sim$ occupation span, number of occupants
 - Artifact size: large artifacts are more likely to interfere.
 - Hazard potential. *e.g.* glass, poop
 - Importance of “other activities”
 - *Higher interference potential means refuse is more likely to be moved farther away.*
- Costs of cleaning up and moving the refuse
 - Scale with distance moved
 - Artifact size: little bits are harder to pick up or sweep than big bits.
 - *Easy cleanup means refuse is more likely to be moved farther away.*

Systemic context processes

Spatial organization discard

- How to recognize cleaning up?
- Ethnoarchaeological evidence: Alyawara men's camp (O'Connell 1987)

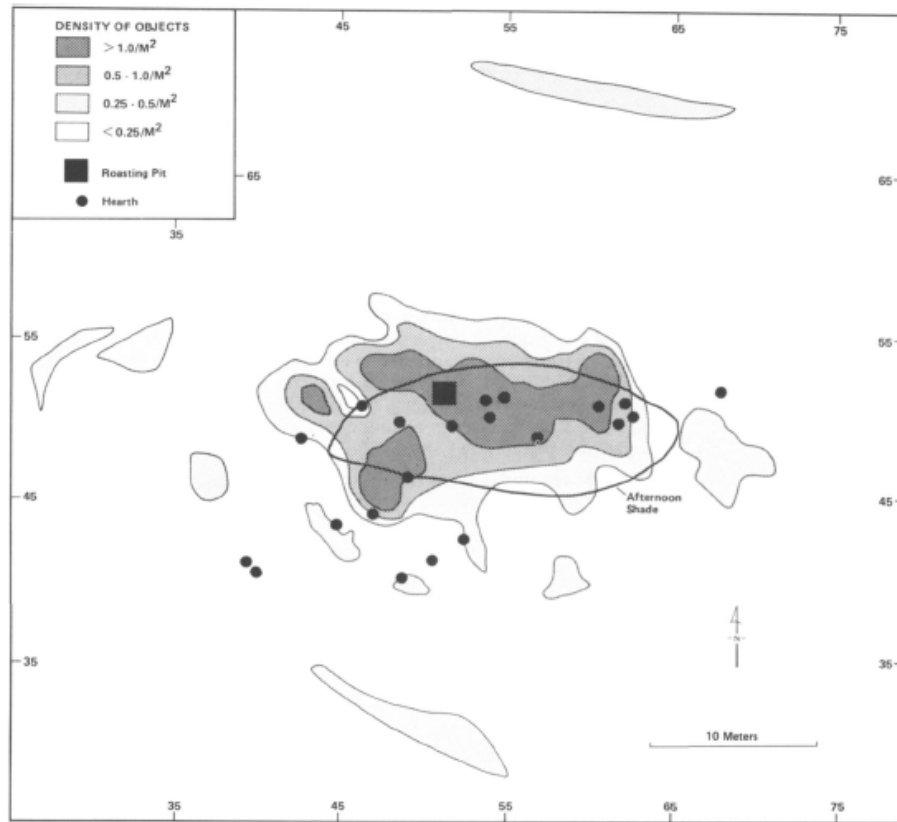


Figure 11. Contour map of refuse density at Bendaijerum *apulla* men's household activity area. Contours computed on refuse distribution data summarized by 2 m square. Total number of refuse items = 972. See Table 7 for description.

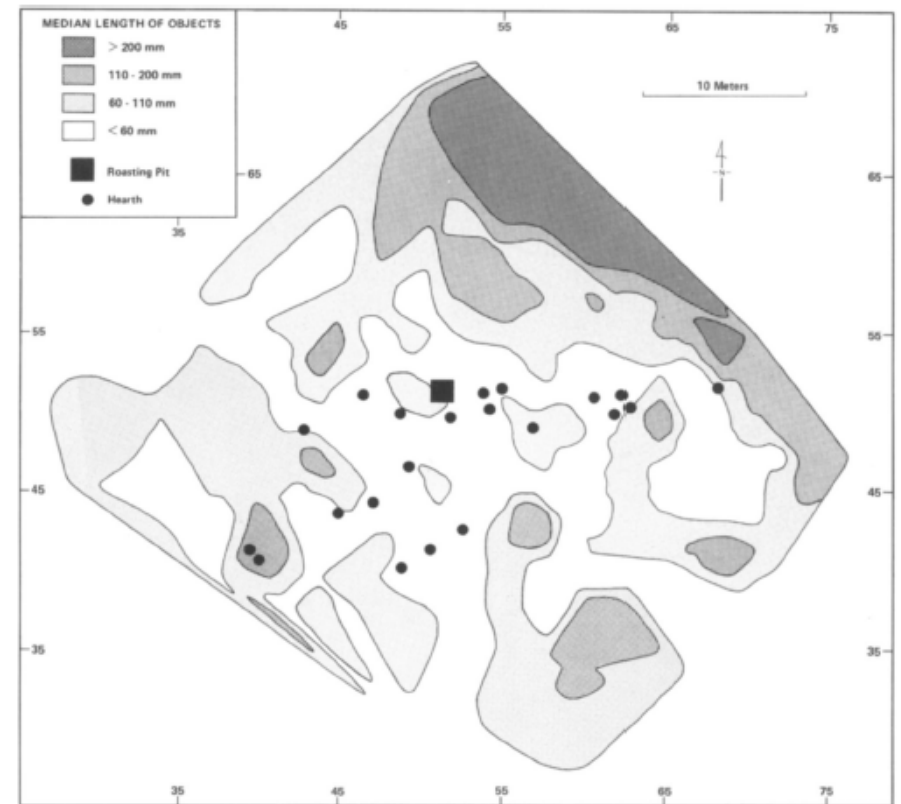


Figure 12. Contour map of median length of non-cloth objects at Bendaijerum *apulla* men's household activity area. Contours computed on data summarized by 2 m square.

Site 8 Example

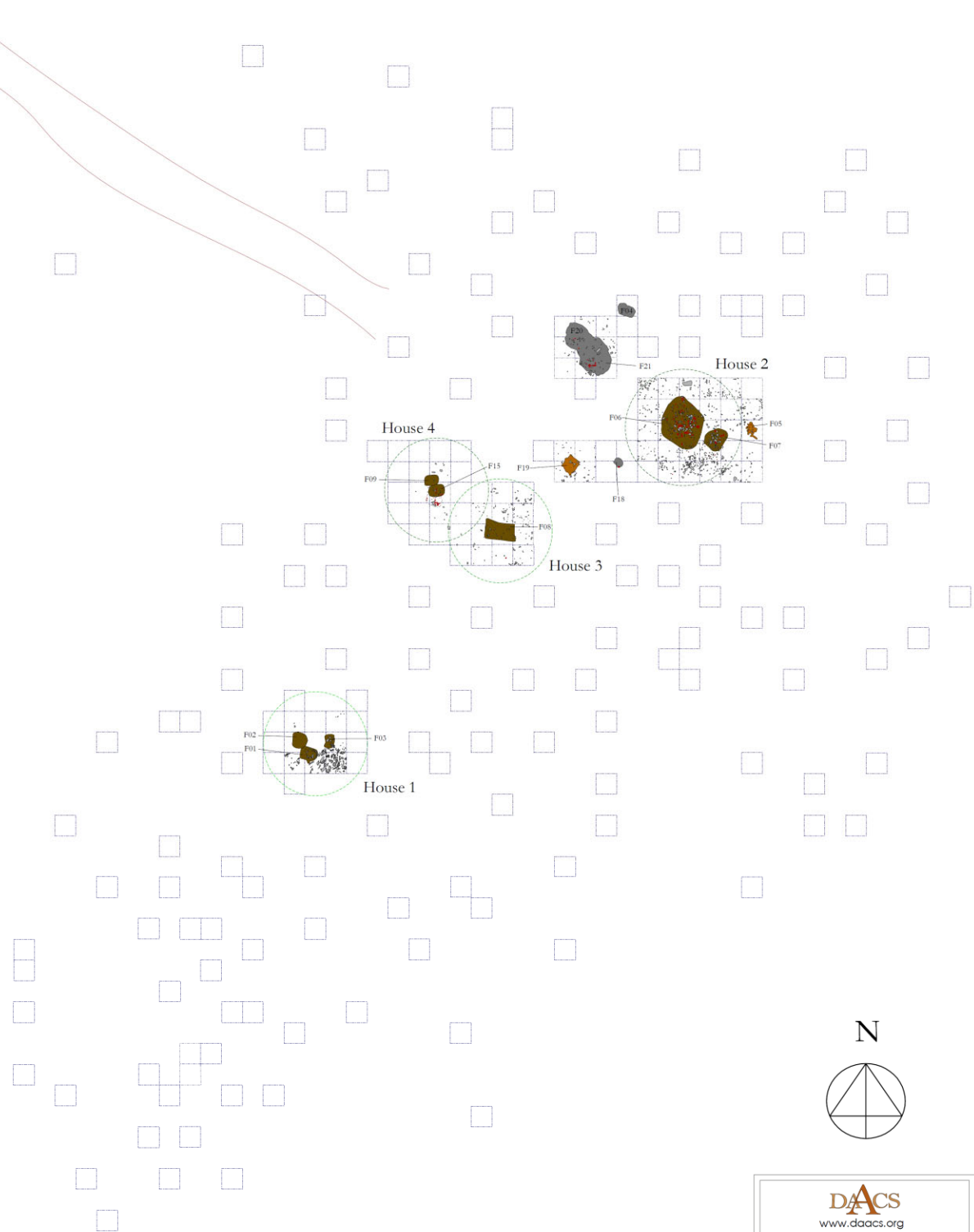
- c. 1770-1800
- heavily plowed
- stratified random sample
- 266 5-foot quadrats

0 25 50



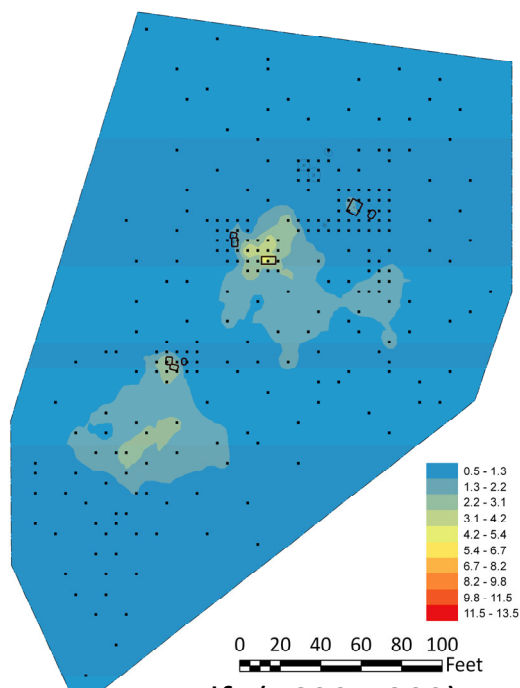
Feet

- = Suspected House Location
- = Subfloor Pit/Cellar
- = Brick
- = Stone
- = Slate
- = Misc. Feature
- = Tree Hole
- = Modern Road

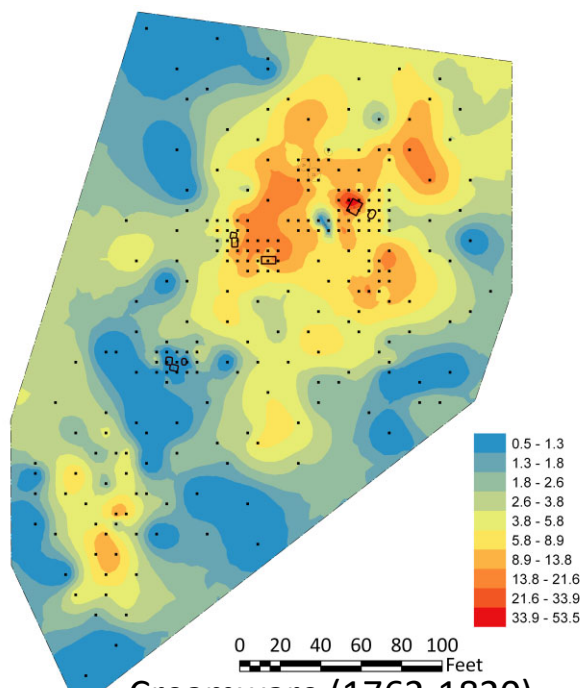


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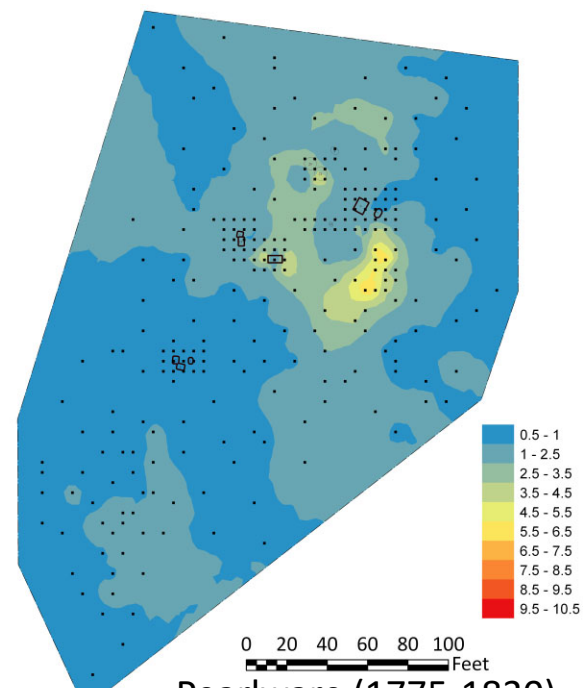
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Date: 04 September 2008



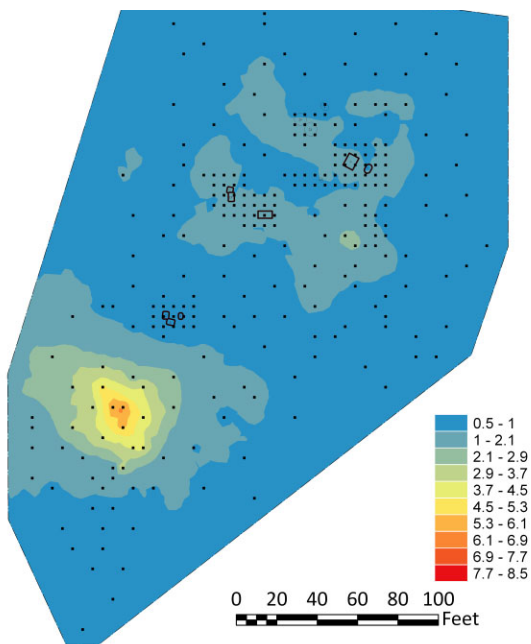
Delft (1600-1802)



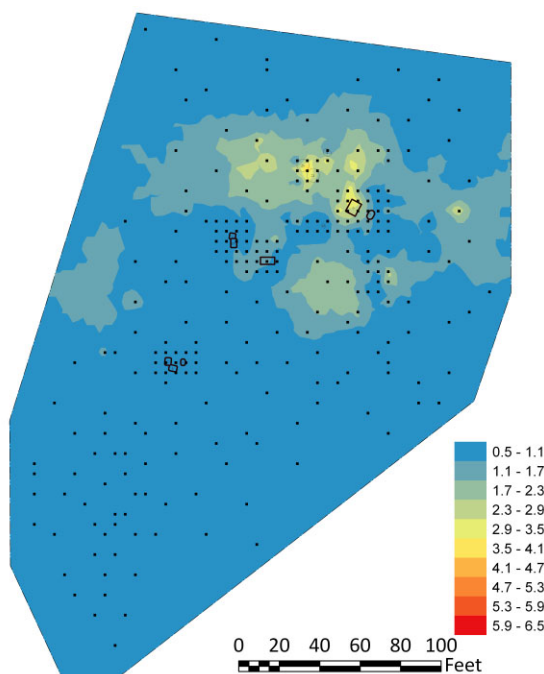
Creamware (1762-1820)



Pearlware (1775-1830)



Redware (post 1600)



Chinese Porcelain (post 1600)

Site 8 (c. 1770-1800)

The Artifact Size Index

A tool to recognize the size sorting that results from cleaning up.

$$ASI_i = \frac{(x_i - pN_i)}{\sqrt{p - (1 - p)N_i}}$$

*Negative ASI: Fewer
Large artifacts than
expected*

*Positive ASI: More
large artifacts than
expected*

p = Site-wide proportion of large artifacts

x_i = Number of **large artifacts** from the i 'th quadrat

N_i = Total artifacts from the i 'th quadrat

So...

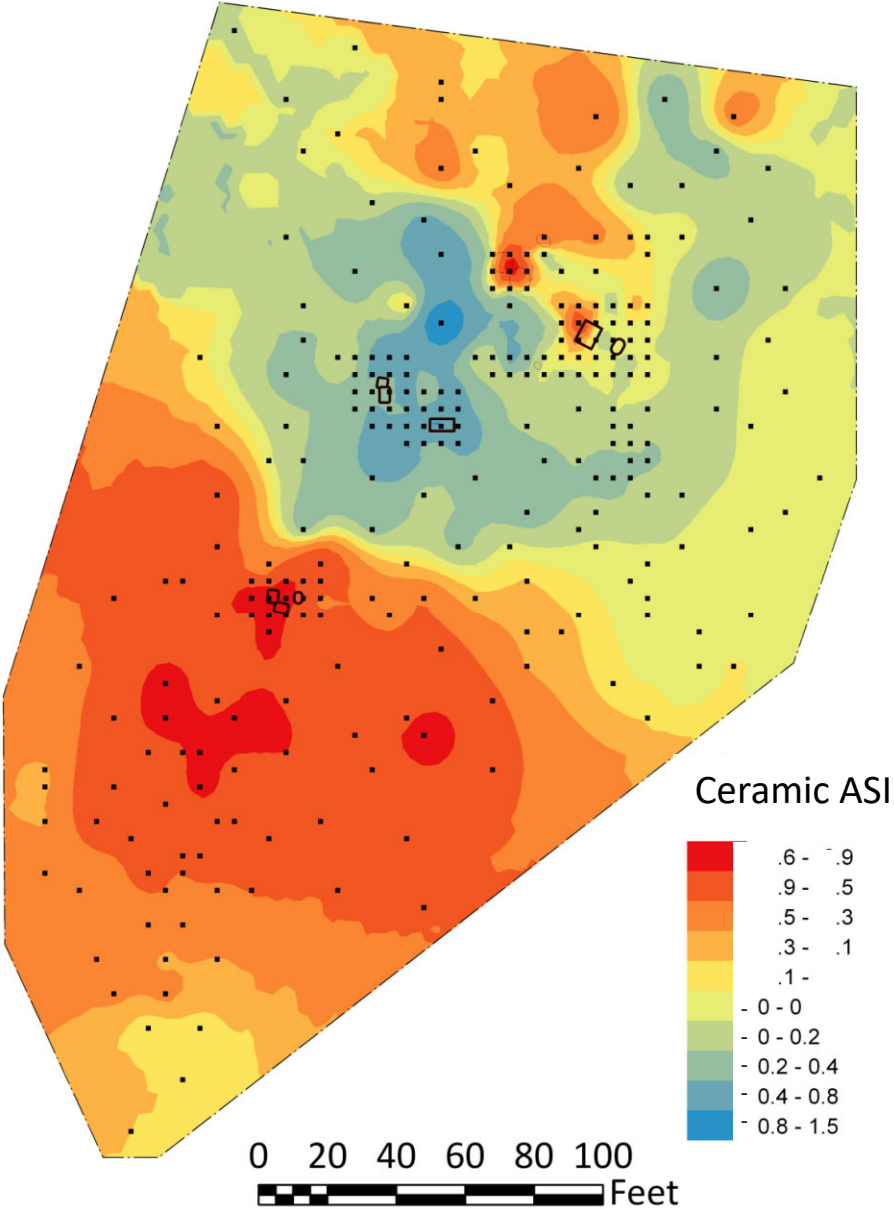
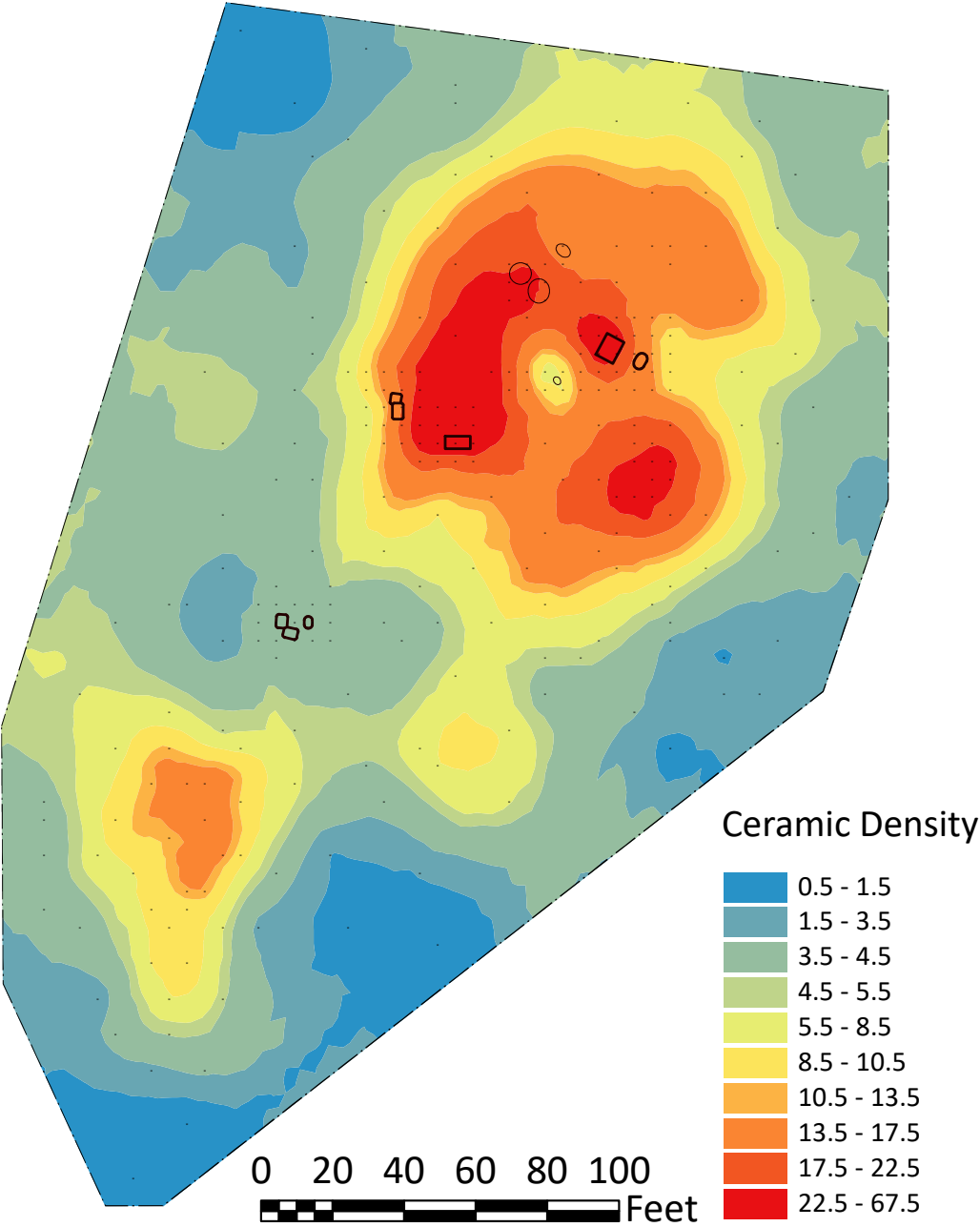
pN_i = Expected number of large artifacts in the i 'th quadrat, if all quads had the site-wide proportion

$\sqrt{p - (1 - p)N_i}$ = Standard deviation of the *Gaussian approximation* to the *binomial distribution*, which is a simple statistical model of the chance of getting 0, 1, 2, 3 ... larger artifacts in a sample of size N_i when the underlying probability of getting a large artifact is p .

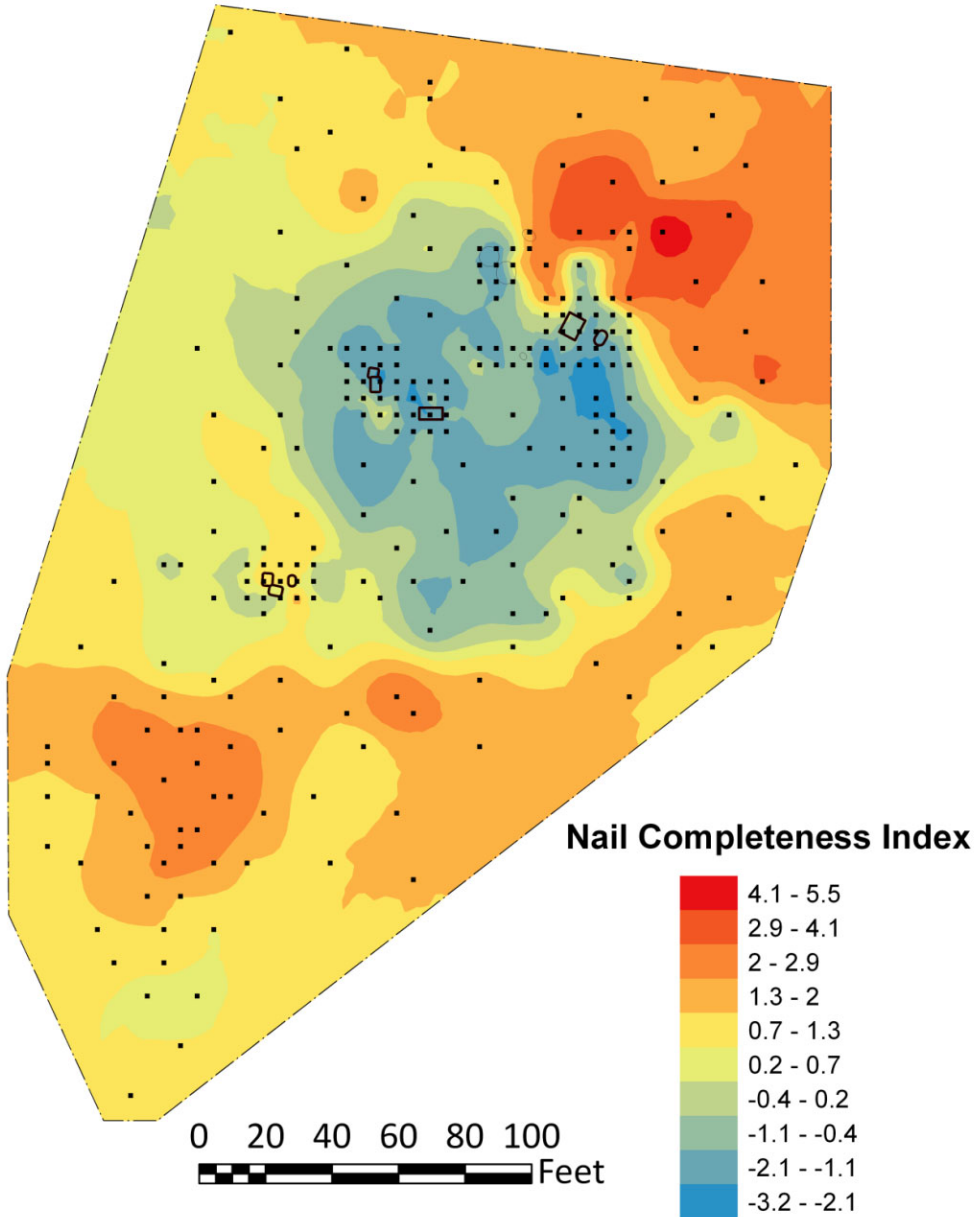
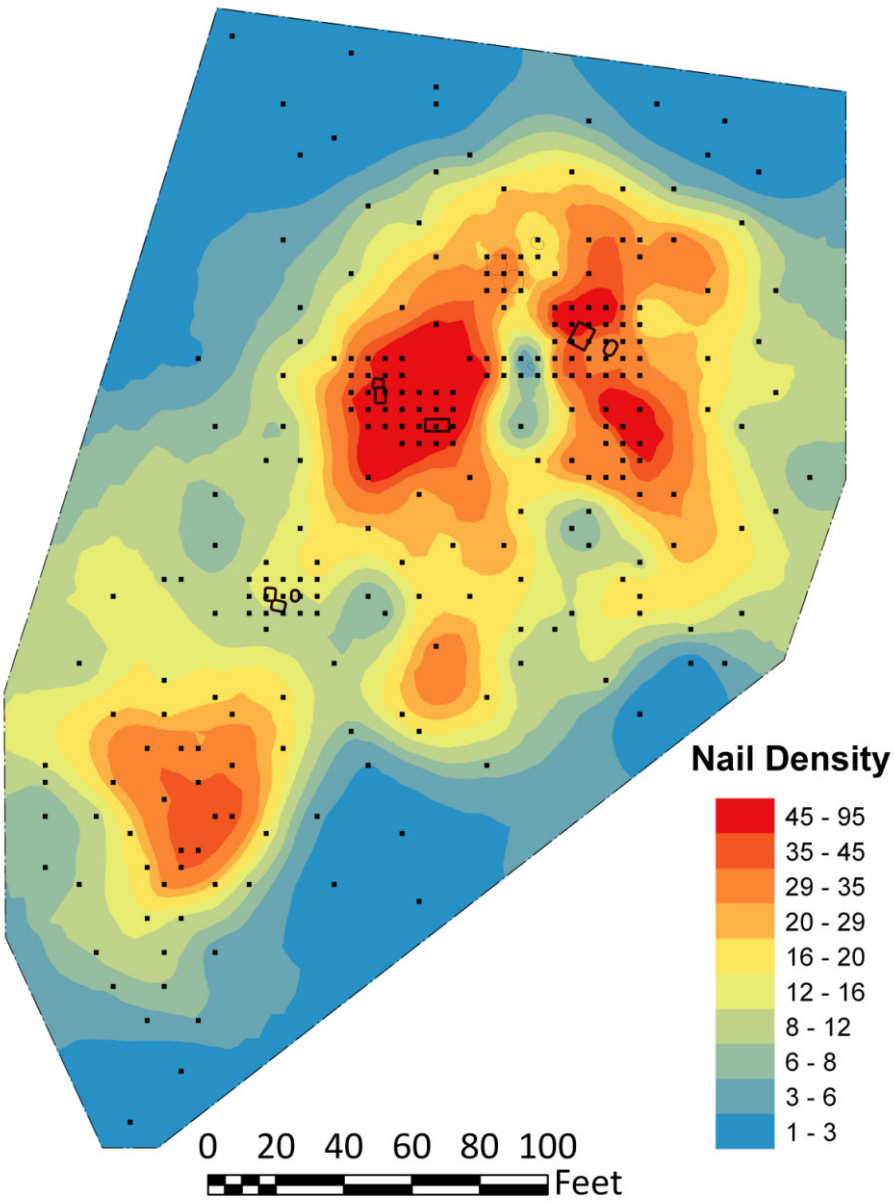
More secret sauce:

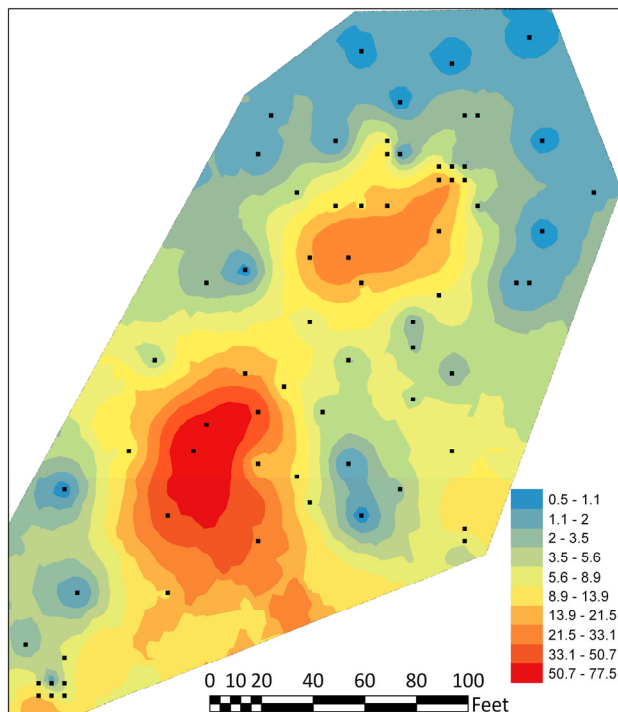
- Interpolate continuous surfaces

Site 8

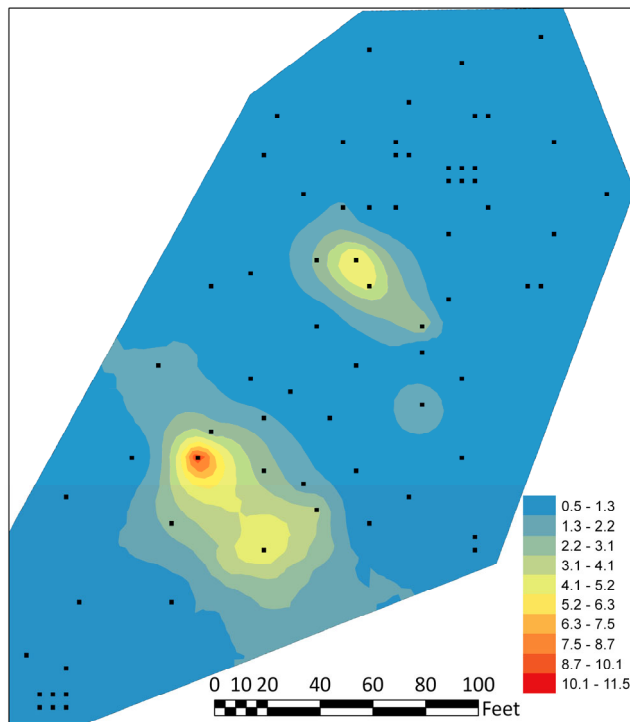


Site 8

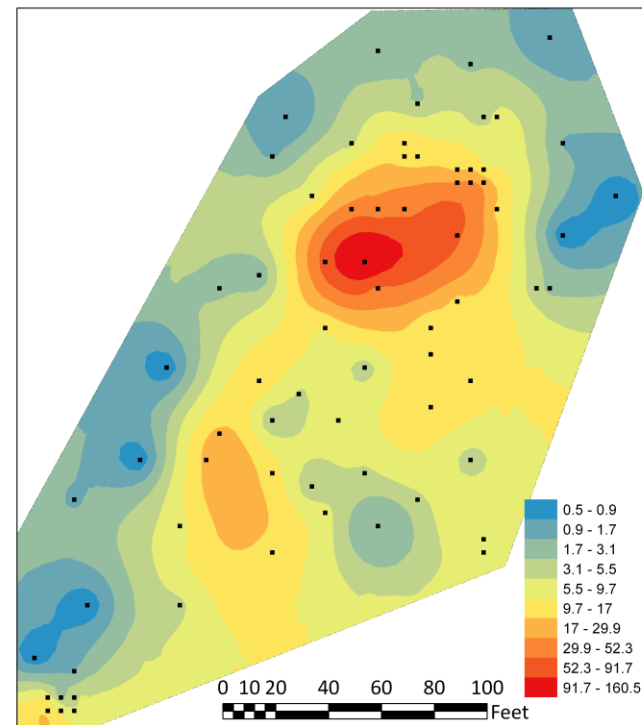




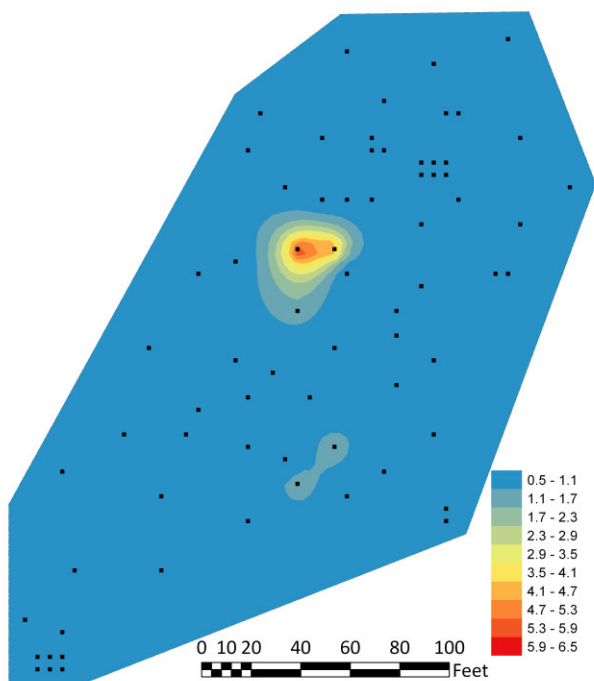
Creamware (1762-1820)



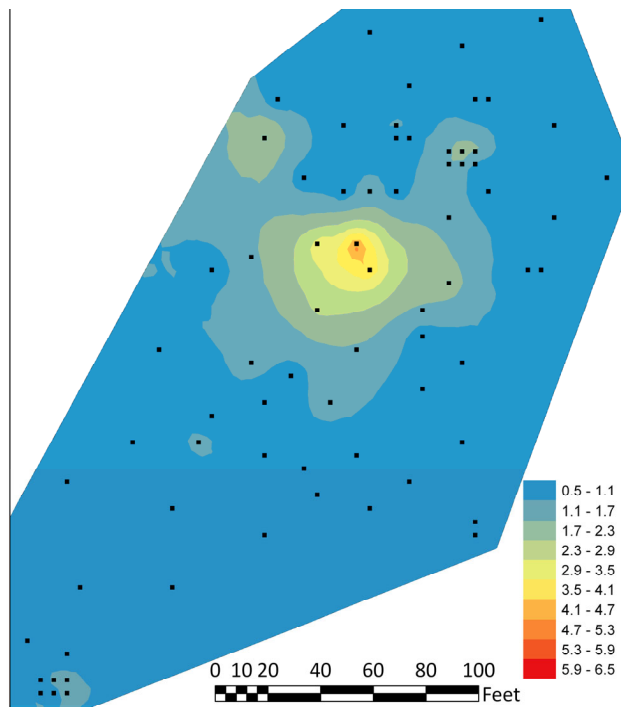
Chinese Porcelain (post 1600)



Pearlware (1775-1830)



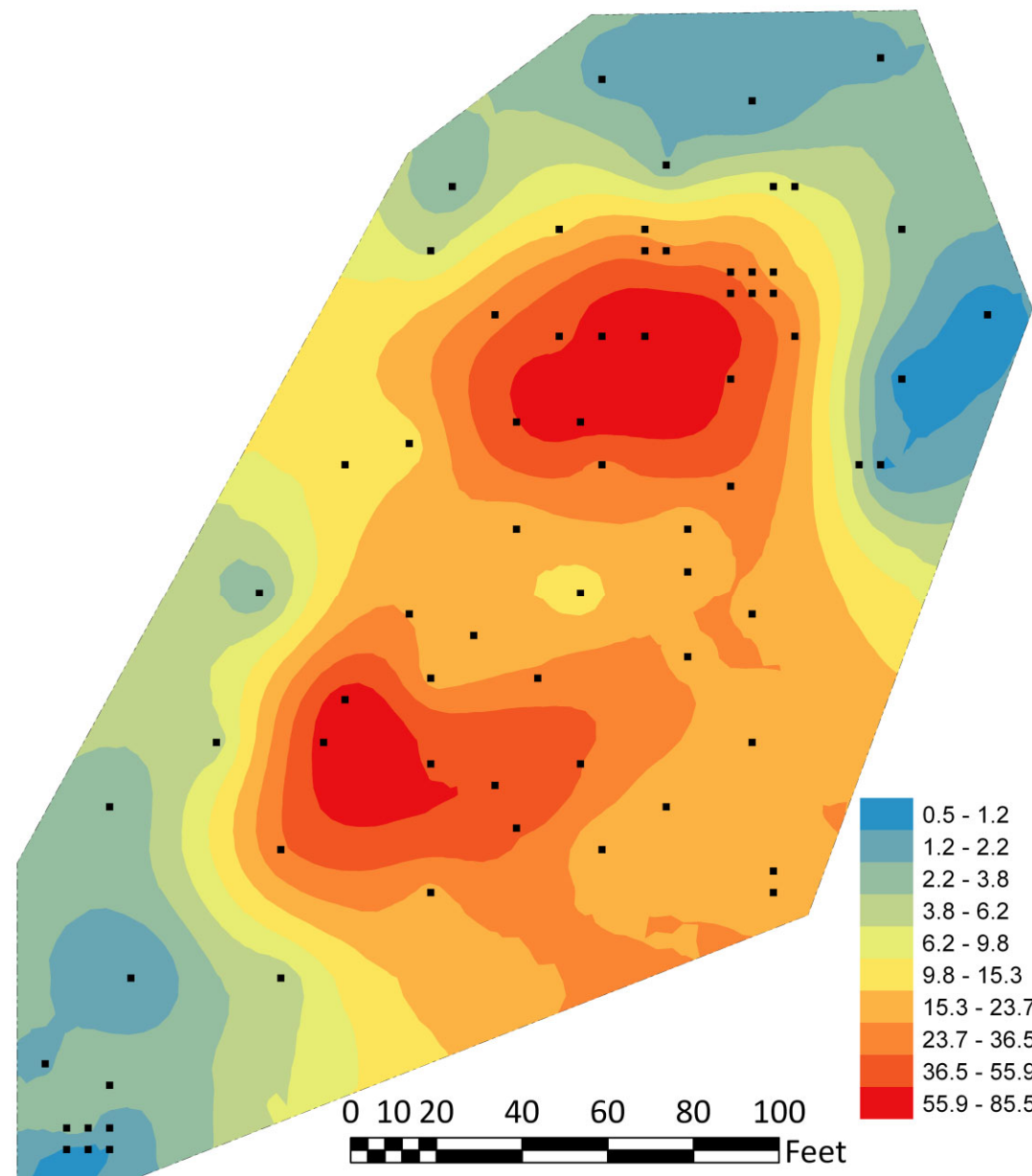
Whiteware



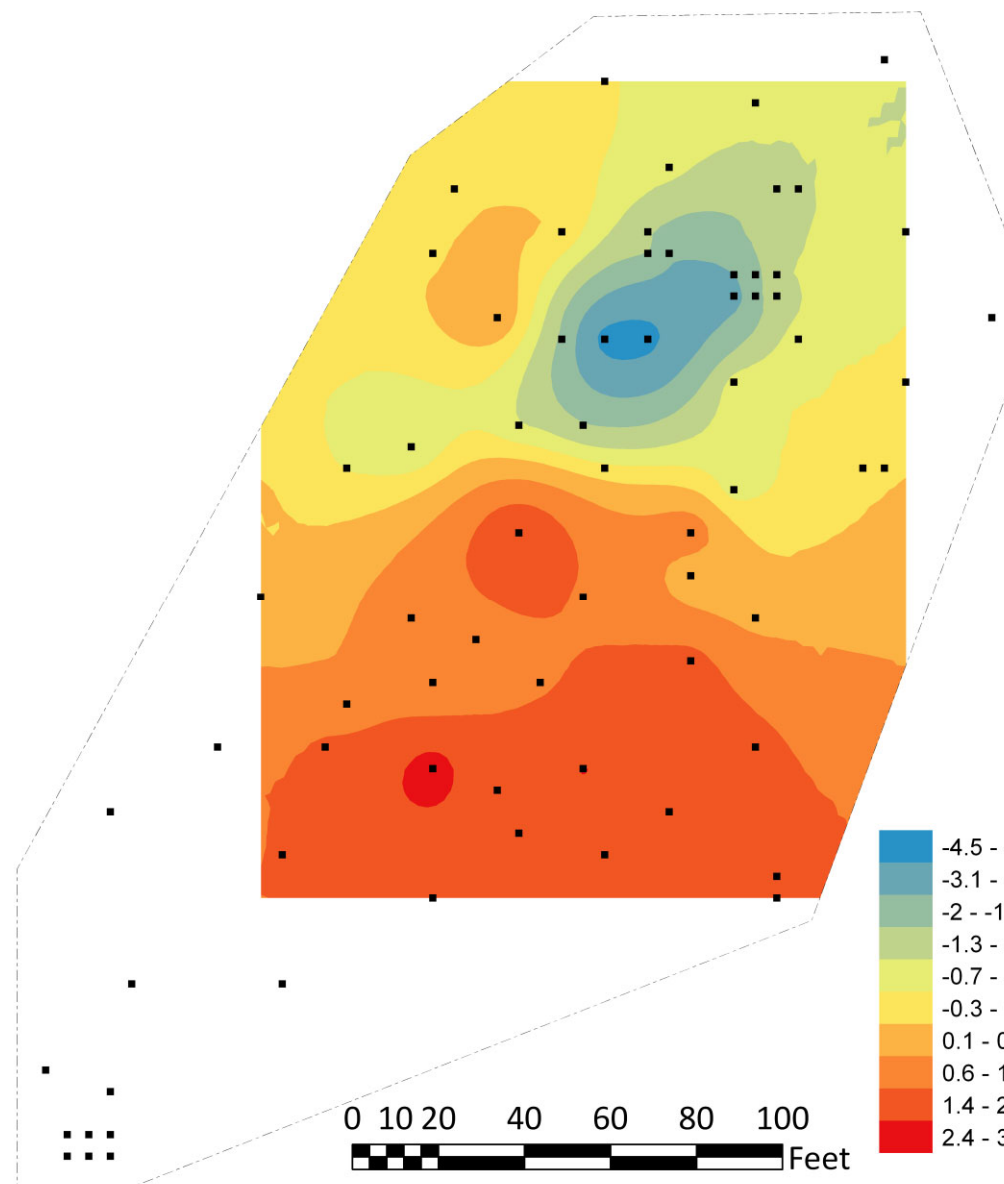
American Stoneware

Site 6

Site 6



Nail Density



Nail Completeness